Superstore Sales Analysis: Final Report

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1. Objective and dataset

The goal of this project is to understand how the Superstore business sells (over time, by product, region, customer segment and shipping mode) and to turn the findings into recommendations. The dataset contains 9,800 order lines (one row per product within an order) covering 3 January 2015 to 30 December 2018, with customer, location, product, ship mode and sales information. **The file has no profit, quantity or discount columns, so every result in this report is about sales, not profitability.**

2. Method and tools

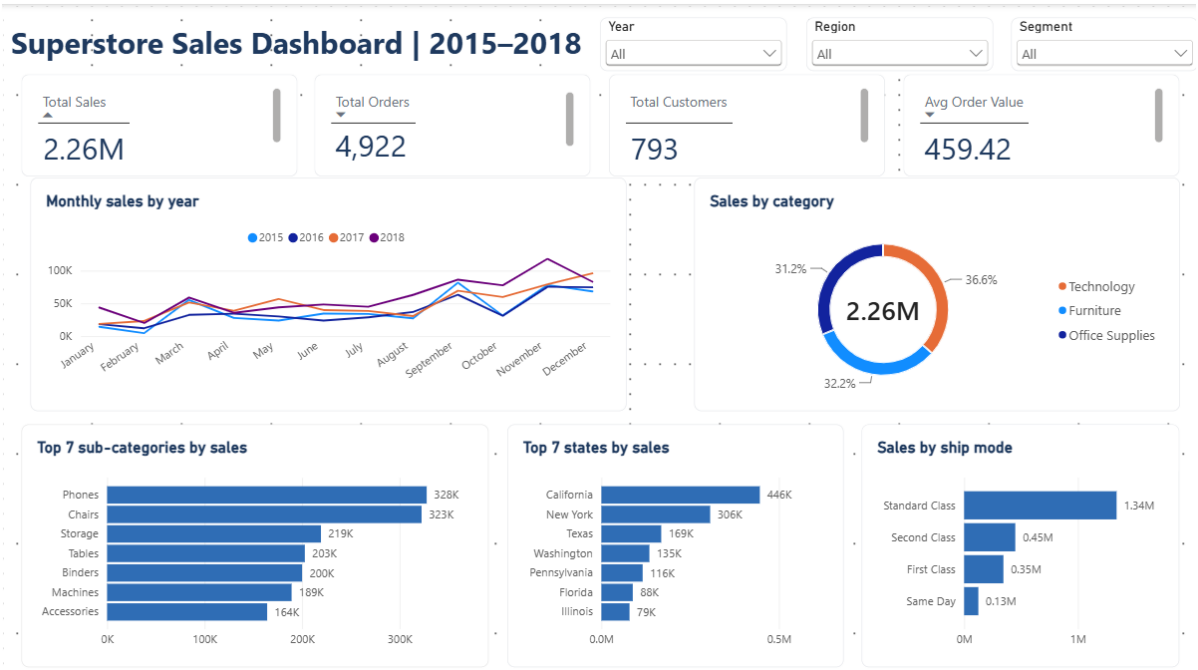
- **Python (pandas):** data understanding, cleaning and exploratory analysis in two notebooks.
- **SQL (SQLite):** 11 queries, including window functions (LAG, RANK, ROW_NUMBER, NTILE, running totals).
- **Power BI:** one interactive dashboard with Year, Region and Segment slicers.
- **Cleaning decisions:** dates converted from day/month/year text; 11 missing postal codes (all Burlington, Vermont) filled with ZIP 05401 and stored as text; 1 duplicate row (identical except Row ID) removed, leaving 9,799 rows; no logical errors found (no ship date before order date, no non-positive sales); year, month, quarter and shipping-days columns added.

3. Key figures

2.26M	4,922	793	459.42
Total sales	Orders	Customers	Average order value

Year	2015	2016	2017	2018
Sales	479,575	459,436	600,193	722,052
Growth	-	-4.2%	+30.6%	+20.3%

4. Dashboard



Power BI dashboard (all filters cleared). The Year, Region and Segment slicers update every chart and card.

5. Insights and recommendations

1. **Growth after a dip in 2016.** Sales fell 4.2% in 2016, then grew 30.6% in 2017 and 20.3% in 2018, reaching about 722K. *Recommendation:* Identify what changed in 2017 (new products, regions or customers) and repeat it.
2. **Strong September to December seasonality.** These four months produce 51.8% of annual sales, with the peak in November (15.5%; November 2018 alone is about 118K). February (2.6%), January (4.2%) and April (6.0%) are the weakest months. *Recommendation:* Build stock, staffing and promotions ahead of September, and run campaigns in the weak months, especially February.
3. **Sales are concentrated in a few products.** Phones (328K) and Chairs (323K) make up 28.8% of sales, while the five lowest sub-categories (Fasteners, Labels, Envelopes, Art and Supplies) contribute only 4.6%. The three categories are almost balanced: Technology 36.6%, Furniture 32.2%, Office Supplies 31.2%. *Recommendation:* Protect availability of the top sellers, and review or bundle the lowest-selling items with them.
4. **Geographic concentration.** California (19.7%) and New York (13.5%) produce 33.3% of sales and the top 10 states 69.1%. West (31%) and East (30%) lead; the South is the weakest region (17%). *Recommendation:* Defend the top states, and test marketing or partnerships in the South and Central regions.
5. **Customer mix and loyalty.** Consumers generate 51% of sales, Corporate 30% and Home Office 19%. Corporate grew fastest, from 128K in 2015 to 236K in 2018. The top 10% of customers give 31% of sales, and customers are very loyal: 98.4% ordered more than once (6.2 orders on average). *Recommendation:* Start a loyalty program for the top customers, and develop the Corporate segment with bulk-order offers.
6. **Shipping preference.** Standard Class handles 59.3% of sales (5.0 days on average); Second Class 3.3 days, First Class 2.2 days and Same Day 0.04 days match their promises. *Recommendation:* Test a free-shipping threshold for Standard Class, or a small premium for faster modes.

6. Validation

The key figures were calculated independently in pandas, SQL and Power BI and agree: total sales 2,261,255.41, 4,922 orders, 793 customers, average order value 459.42, yearly growth of -4.2%, +30.6% and +20.3%, and the top 10 states at 69.1% of sales. One expected difference: the top 10% of customers is 31.2% in SQL (NTILE puts 80 customers in the first group) and 30.9% in pandas (79 customers); this report uses 31%.

7. Limitations and next steps

- No profit, quantity or discount data: the analysis shows what sells most, not what is most profitable. Adding profit would be the first improvement.
- No customer demographics or marketing data, so the reasons behind the 2016 dip and the 2017 recovery cannot be explained from this file.
- Next steps: add profit analysis if available, forecast the 2019 seasonal pattern, and build customer segments (for example by order frequency and value).

8. Project files

Code, notebooks, SQL queries, dashboard file and screenshots are in the GitHub repository: **superstore-sales-analysis** (folders: data, notebooks, sql, dashboard, report, images).